

**PRE-MEDICAL : ENTHUSIAST ADVANCE COURSE PHASE : MED-I (B)**

IMPORTANT NOTE : Students having 8 digits **Form No.** must fill two zero before their **Form No.** in OMR. For example, if your **Form No.** is 12345678, then you have to fill **0012345678**.

Test Booklet Code

This Booklet contains **28** pages.**E6**

Do not open this Test Booklet until you are asked to do so.

Important Instructions :

- The Answer Sheet is inside this Test Booklet. When you are directed to open the Test Booklet, take out the Answer Sheet and fill in the particulars on ORIGINAL Copy carefully with **blue/black** ball point pen only.
- The test is of **3 hours 20 minutes** duration and the Test Booklet contains **200** multiple-choice questions (four options with a single correct answer) from **Physics, Chemistry and Biology (Biology-I and Biology-II)**. **50** questions in each subject are divided into **two Sections (A and B)** as per details given below :
 - Section A** shall consist of **35 (Thirty-five)** Questions in each subject (Question Nos - 1 to 35, 51 to 85, 101 to 135 and 151 to 185). All questions are compulsory.
 - Section B** shall consist of **15 (Fifteen)** questions in each subject (Question Nos - 36 to 50, 86 to 100, 136 to 150 and 186 to 200). In Section B, a candidate needs to **attempt any 10 (Ten)** questions out of **15 (Fifteen)** in each subject. **Candidates are advised to read all 15 questions in each subject of Section B** before they start attempting the question paper. In the event of a candidate attempting more than ten questions, **the first ten questions answered by the candidate shall be evaluated.**
- Each question carries **4** marks. For each correct response, the candidate will get **4** marks. For each incorrect response, **one mark** will be deducted from the total scores. **The maximum marks are 720.**
- Use **Blue/Black Ball Point Pen only** for writing particulars on this page/markings responses on Answer Sheet.
- Rough work is to be done in the space provided for this purpose in the Test Booklet only.
- On completion of the test, the candidate **must hand over the Answer Sheet (ORIGINAL and OFFICE Copy) to the Invigilator** before leaving the Room/Hall. The candidates are allowed to take away this Test Booklet with them.
- The candidates should ensure that the Answer Sheet is not folded. Do not make any stray marks on the Answer Sheet. Do not write your Form No. anywhere else except in the specified space in the Test Booklet/Answer Sheet.
- Use of white fluid for correction is **NOT** permissible on the Answer Sheet.
- Each candidate must show on-demand his/her Allen ID Card to the Invigilator.
- No candidate, without special permission of the Invigilator, would leave his/her seat.
- The candidates should not leave the Examination Hall without handing over their Answer Sheet to the Invigilator on duty and sign (with time) the Attendance Sheet **twice**. **Cases, where a candidate has not signed the Attendance Sheet second time, will be deemed not to have handed over the Answer Sheet and dealt with as an Unfair Means case.**
- Use of Electronic/Manual Calculator is prohibited.
- The candidates are governed by all Rules and Regulations of the examination with regard to their conduct in the Examination Room/Hall. All cases of unfair means will be dealt with as per the Rules and Regulations of this examination.
- No part of the Test Booklet and Answer Sheet shall be detached under any circumstances.**
- The candidates will write the Correct Test Booklet Code as given in the Test Booklet/Answer Sheet in the Attendance Sheet.
- Compensatory time of one hour five minutes will be provided for the examination of three hours and 20 minutes duration, whether such candidate (having a physical limitation to write) uses the facility of scribe or not.

Name of the Candidate (in Capitals) : _____

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Centre of Examination (in Capitals) : _____

Candidate's Signature : _____ Invigilator's Signature : _____

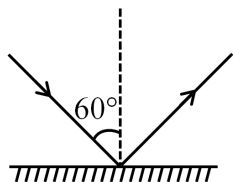
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Your Target is to secure Good Rank in Pre-Medical 2024

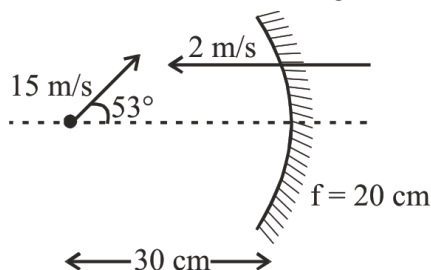
Topic : THERMAL PHYSICS, MAGNETIC EFFECT OF CURRENT [BACK UNIT] : RAY OPTICS, AND OPTICAL INSTRUMENT

SECTION-A (PHYSICS)

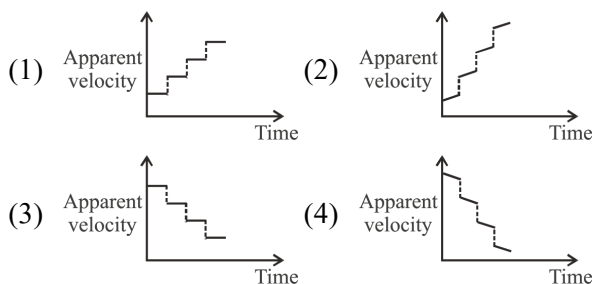
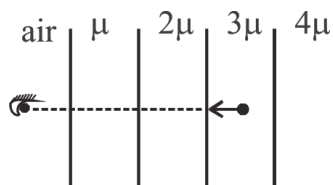
1. At what angle mirror should be rotated to obtained reflected ray along +x axis.



- (1) 30° (2) 15°
(3) 45° (4) 60°
2. Find the velocity of image w.r.t. ground in a situation as shown in the figure :-



- (1) $-44\hat{i} - 24\hat{j}$ (2) $-46\hat{i} - 24\hat{j}$
(3) $-42\hat{i} - 12\hat{j}$ (4) $+46\hat{i} + 24\hat{j}$
3. A particle is moving with a real velocity V m/sec. along the straight line shown. An observer at the end of the same line is viewing the particle. Which of the following graphs most appropriately represent the velocity of the particle as seen by the observer ? All the medium boundaries are in state of rest :



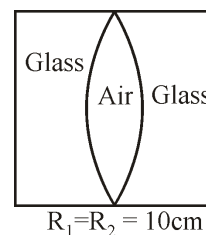
4. Width of a slab is 6 cm whose $\mu = \frac{3}{2}$. If its rear surface is silvered and object is placed at a distance 28 cm from the front face. Calculate the final position of the image from the silvered surface.

- (1) 32 cm (2) 30 cm
(3) 26 cm (4) None

5. A vessel is half filled with a liquid of refractive index μ . The other half of the vessel is filled with an immiscible liquid of refractive index 2μ . The apparent depth of the vessel is 25% of actual depth. The value of μ is :-

- (1) $\frac{1}{3}$ (2) $\frac{3}{1}$ (3) $\frac{4}{3}$ (4) $\frac{3}{4}$

6. In given situation focal length and nature of lens will be



- (1) 10 cm, converging
(2) 10 cm, diverging
(3) ∞ , Neither converging nor diverging
(4) 0, Concave
7. A man can see only between 75 cm and 200 cm. The power of lens to correct the near point will be
- (1) $+8/3$ D (2) $+3$ D
(3) -3 D (4) $-8/3$ D
8. In a compound microscope, the focal lengths of two lenses are 1.5 cm and 6.25 cm. An object is placed at 2 cm from objective and the final image is formed at 25 cm from eye lens. The distance between the two lenses is
- (1) 6.00 cm (2) 7.75 cm
(3) 9.25 cm (4) 11.00 cm

9. A telescope consisting of an objective of focal length 60 cm and an eyepiece of focal length 5 cm is focussed to a distant object in such a way that parallel rays emerge from the eye piece. If the object subtends an angle of 2° at the objective, then find the angular width of the image.

- (1) 12° (2) 6°
(3) 24° (4) None

10. It is necessary to illuminate the bottom of a well by reflected solar beam when the light is incident at an angle 40° to the vertical. At what angle to the horizontal should a plane mirror be placed?

- (1) 65° (2) 70° (3) 75° (4) 80°

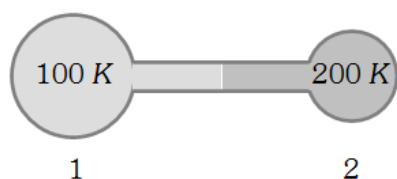
11. Two thin lenses when put in contact produce a combination of power +10 dioptre. When they are 25 cm apart, power reduces to +6 dioptre. Focal lengths of the lenses are

- (1) 12.5 cm, 50 cm
(2) 12.5 cm, 25 cm
(3) 25 cm, 50 cm
(4) 50 cm, 30 cm

12. A gas contained in a box of 0.1 m^3 at atmospheric pressure is connected at constant temperature to another vessel of 0.09 m^3 . Consequent change in pressure is X metre of Hg. Then X is -

- (1) 0.4 (2) 0.5 (3) 0.36 (4) 0.3

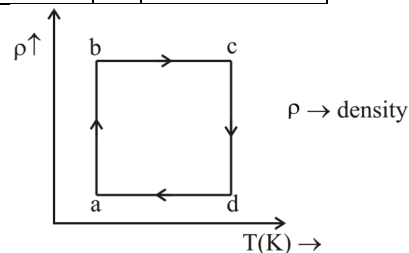
13. Figure shows two flasks connected to each other. The volume of the flask 1 is twice that of flask 2. The system is filled with an ideal gas at temperature 100 K and 200 K respectively. If the mass of the gas in 1 be m then what is the mass of the gas in flask 2



- (1) m (2) $m/2$ (3) $m/4$ (4) $m/8$

14. In the $p - T$ graph shown in the figure, match the following column :

	Column-I		Column-II
(A)	Process ab	(p)	isochoric process
(B)	Process bc	(q)	$\Delta U = 0$
(C)	Process cd	(r)	ρ increasing
(D)	Process da	(s)	ρ decreasing

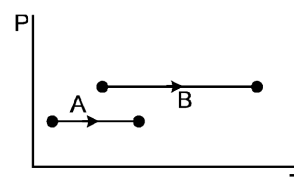


- (1) $A \rightarrow q, s; B \rightarrow p, q; C \rightarrow p, s; D \rightarrow r, q$
(2) $A \rightarrow q, r; B \rightarrow p; C \rightarrow q, s; D \rightarrow p$
(3) $A \rightarrow p, s; B \rightarrow q, r; C \rightarrow r, s; D \rightarrow s, r$
(4) $A \rightarrow p, q; B \rightarrow q, r; C \rightarrow r, s; D \rightarrow s, p$

15. n molecules of an ideal gas are enclosed in cubical box at temperature T and pressure P. If the number of molecules in the box is trippled then new temperature and pressure become T' and P' respectively, but the total energy of gas system remains unchanged, then -

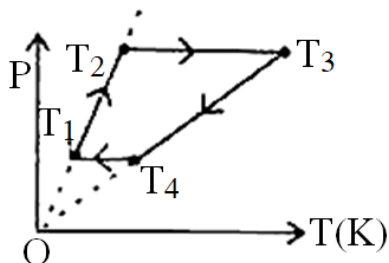
- (1) $P = P'$ and $T = T'$
(2) $P = 3P'$ and $T' = \frac{1}{3}T$
(3) $P' = 3P$ and $T' = T$
(4) $P' = P$ and $T' = \frac{T}{3}$

16. Consider two processes on a system as shown in fig. The volumes in the initial states are the same in the two processes and the volumes in the final states are also the same. Let ΔW_1 and ΔW_2 be the work done by the system in the processes A and B respectively.

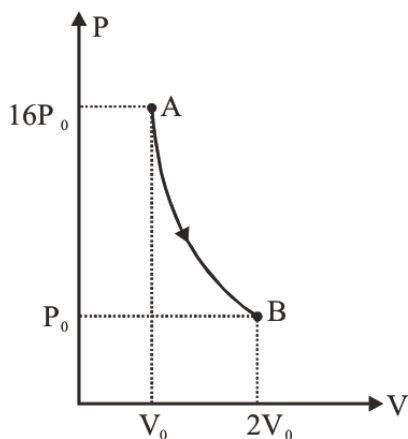


- (1) $\Delta W_1 > \Delta W_2$
(2) $\Delta W_1 = \Delta W_2$
(3) $\Delta W_1 < \Delta W_2$
(4) Nothing can be said about the relation between ΔW_1 and ΔW_2

17. Two moles of an ideal monoatomic gas performs a cyclic process as shown in the figure. The temperatures in different states are $T_1 = 400$ K, $T_2 = 800$ K, $T_3 = 2400$ K and $T_4 = 1200$ K. What is work done by the gas during the cycle?

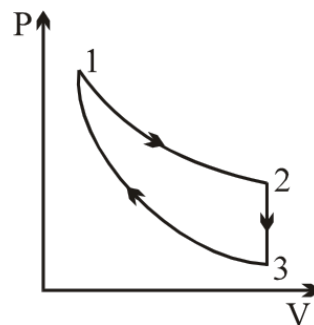


- (1) 16.35 kJ (2) 32.45 kJ
(3) 13.3 kJ (4) 10.95 kJ
18. One mole of an ideal gas at an initial temperature of T K does $6R$ joules of work adiabatically. If the ratio of specific heats of this gas at constant pressure and at constant volume is $5/3$, the final temperature of gas will be :
- (1) $(T + 2.4)K$ (2) $(T - 2.4)K$
(3) $(T + 4)K$ (4) $(T - 4)K$
19. Figure demonstrates a polytropic process for an ideal gas. The work done by the gas in process AB is :-

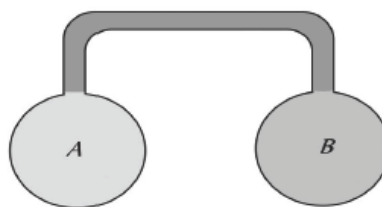


- (1) $\frac{15}{2}P_0V_0$
(2) $\frac{14}{3}P_0V_0$
(3) $8P_0V_0$
(4) Insufficient information

20. Three processes form a thermodynamic cycle as shown on P-V diagram for an ideal gas. Process $1 \rightarrow 2$ takes place at constant temperature (300K). Process $2 \rightarrow 3$ takes place at constant volume. During this process 40J of heat leaves the system. Process $3 \rightarrow 1$ is adiabatic and temperature T_3 is 275K. Work done during the process $3 \rightarrow 1$ is :-



- (1) -40J (2) -20J (3) +40J (4) +20J
21. Two spherical vessel of equal volume, are connected by a narrow tube. The apparatus contains an ideal gas at one atmosphere and 300K. Now if one vessel is immersed in a bath of constant temperature 600K and the other in a bath of constant temperature 300 K. Then the common pressure will be



- (1) 1 atm (2) $\frac{4}{5}$ atm
(3) $\frac{4}{3}$ atm (4) $\frac{3}{4}$ atm
22. A container encloses 32 gm of O_2 at $127^\circ C$ and pressure P_0 . A small hole is made in the container and the gas leaks out. At a certain state, pressure of gas becomes $\frac{P_0}{2}$ and its temperature $27^\circ C$. Mass of the gas that has leaked out till this stage is nearly :-
- (1) 6.3 gm (2) 7.4 gm
(3) 16.3 gm (4) 10.7 gm

23. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : Ampere's circuital law is not suitable for a finite straight wire carrying current I .

Reason (R) : One cannot calculate magnetic field due to a finite straight current carrying wire by using Biot Savart Law.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
 (2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
 (3) **(A)** is true but **(R)** is false.
 (4) **(A)** is false but **(R)** is true.
24. In a region a uniform magnetic field acts in horizontal plane towards north. If cosmic particles (80% protons) falling vertically downwards, then they are deflected towards –
- (1) North (2) South
 (3) East (4) West
25. A solid cylindrical wire of radius ' R ' carries a current ' I '. The magnetic field is $5\mu\text{T}$ at a point, which is ' $2R$ ' distance away from the axis of wire. Magnetic field at a point which is $R/3$ distance inside from the surface of the wire is :-
- (1) $\frac{10}{3}\mu\text{T}$ (2) $\frac{20}{3}\mu\text{T}$
 (3) $\frac{5}{3}\mu\text{T}$ (4) $\frac{40}{3}\mu\text{T}$
26. A solenoid has length 0.4m , radius 1cm and 400 turns of wire. If a current of 5A is passed through this solenoid, what is the magnetic field inside the solenoid ?

- (1) $6.28 \times 10^{-4}\text{T}$
 (2) $6.28 \times 10^{-3}\text{T}$
 (3) $6.28 \times 10^{-7}\text{T}$
 (4) $6.28 \times 10^{-6}\text{T}$

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PHASE - MED

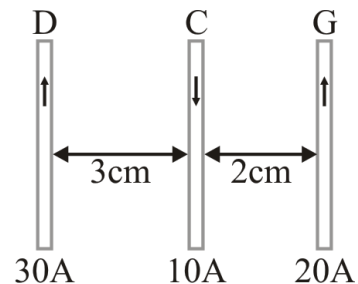
27. A proton, a deuteron and an alpha particle are moving with the same momentum and enter in a uniform transverse magnetic field. The ratio of the radii of their circular paths will be –

- (1) $1 : 1 : 1$ (2) $1 : 2 : 2$
 (3) $2 : 1 : 1$ (4) $2 : 2 : 1$

28. A particle of mass ' m ' and charge ' Q ' is accelerated in cyclotron. If magnetic field is ' B ' and radius of cyclotron is r then find the kinetic energy of the particle.

- (1) $\frac{QBr}{2m}$ (2) $\frac{Q^2B^2r}{2m}$
 (3) $\frac{Q^2B^2r^2}{2m}$ (4) $\frac{Q^2B^2r^2}{m}$

29. Three long, straight parallel wires carrying current, are arranged as shown in figure. The force experienced by a 25cm length of wire C is :-



- (1) 10N
 (2) $2.5 \times 10\text{N}$
 (3) Zero
 (4) $1.5 \times 10\text{N}$
30. A uniform conducting wire of length $12a$ and resistance ' R ' is wound up as a current carrying coil in the shape of,
 (i) an equilateral triangle of side ' a '.
 (ii) a square of side ' a '.
 The magnetic dipole moments of the coil in each case respectively are:
- (1) $\sqrt{3}Ia^2$ and $3Ia^2$ (2) $3Ia^2$ and Ia^2
 (3) $3Ia^2$ and $4Ia^2$ (4) $4Ia^2$ and $3Ia^2$

31. A ring of radius R , made of an insulating material carries a charge Q uniformly distributed on it. If the ring rotates about the axis passing through its centre and normal to plane of the ring with constant angular speed ω , then the magnitude of the magnetic moment of the ring is :-

- (1) $Q\omega R^2$ (2) $\frac{1}{2} Q\omega R^2$
(3) $Q\omega^2 R$ (4) $\frac{1}{2} Q\omega^2 R$

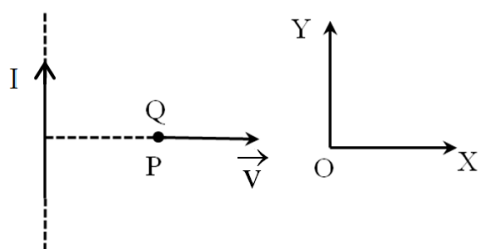
32. A proton is moving with velocity $\vec{v} = (3\hat{i} + 2\hat{j})$ m/s in a magnetic field $\vec{B} = (3\hat{i} + 2\hat{j})$ T. The magnetic force on proton is.

- (1) $1.6 \times 10^{-19} (6\hat{i} + 4\hat{j})$ N
(2) $6 \times 10^{-19} \hat{k}$ N
(3) $(4\hat{i} + 6\hat{j}) \times 10^{-19}$ T
(4) Zero

33. A magnetic dipole of dipole moment $10(\hat{i} + \hat{j} + \hat{k})$ A-m² is placed in magnetic field $(0.6\hat{i} + 0.4\hat{j} + 0.5\hat{k})$ Tesla, force acting on dipole is :

- (1) $(\hat{i} + \hat{j} - 2\hat{k})$ N (2) $(\hat{i} + \hat{j} + 2\hat{k})$ N
(3) $(\hat{i} - \hat{j} + 2\hat{k})$ N (4) Zero

34. A very long straight wire carries a current I . At the instant when a charge $+Q$ at point P has velocity $\vec{v}$, as shown the magnetic force on the charge is



- (1) Along OX (2) Opposite to OY
(3) Along OY (4) Opposite to OX

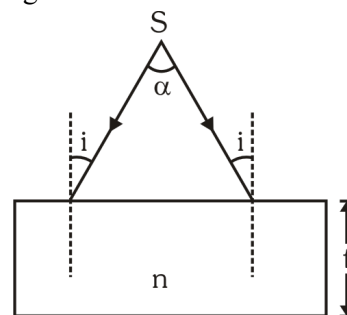
35. Cyclotron is used to accelerate

- (1) Electrons (2) Neutrons
(3) Positive ions (4) Negative ions

36. An object is kept between a plane mirror and a concave mirror facing each other. The distance between the mirrors is 30 cm. The radius of curvature of the concave mirror is 40 cm. What should be the distance of the object from the plane mirror so that after two successive reflections the final image is formed on the object itself : [consider first reflection on plane mirror]

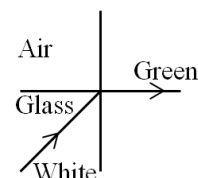
- (1) 5 cm (2) 20 cm
(3) 10 cm (4) Not possible

37. A divergent beam of light from a point source S having divergence angle α falls symmetrically on a glass slab as shown in the figure. The angles of incidence of the two extreme rays are equal. If the thickness of the glass slab is t and its refractive index is n , then the divergence angle of the emergent beam is



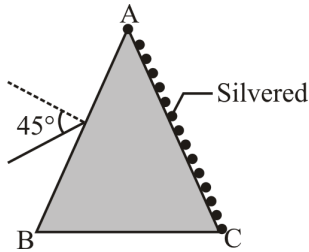
- (1) Zero (2) α
(3) $\sin^{-1} \left(\frac{1}{n} \right)$ (4) $2\sin^{-1} \left(\frac{1}{n} \right)$

38. White light is incident on the interface of glass and air as shown in the figure. If green light is just totally internally reflected then the emerging ray in air contains:

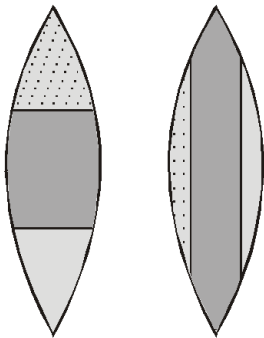


- (1) yellow, orange, red
(2) violet, indigo, blue
(3) all colours
(4) all colours except green

39. A prism ABC of angle 30° has its face AC silvered. A ray of light incident at an angle of 45° at the face AB retraces its path after refraction at face AB and reflection at face AC. The refractive index of the material of the prism is :-

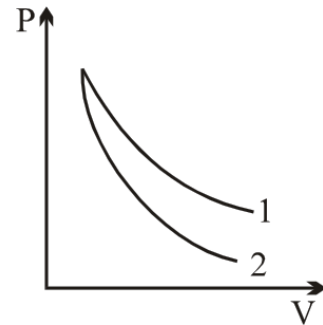


- (1) 1.5 (2) $\frac{3}{\sqrt{2}}$
 (3) $\sqrt{2}$ (4) $\frac{4}{3}$
40. Figure shows two convex lenses A and B, each made up of three different transparent materials. The number of images formed, of an object kept on the principal axis of each lens will be



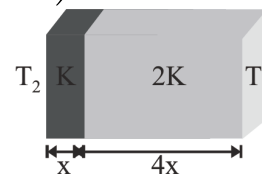
- (1) 3 and 3
 (2) 3 and 1
 (3) 1 and 1
 (4) 3 and 2
41. If the pressure of an ideal gas is increased to four times isothermally, its rms speed will become n times, where n is
- (1) 4
 (2) 2
 (3) $\frac{1}{2}$
 (4) 1

42. P-V plots for two gases during adiabatic processes are shown in the figure. Plots 1 and 2 should correspond respectively to



- (1) He and O₂ (2) O₂ and He
 (3) He and Ar (4) O₂ and N₂
43. In a Carnot engine, if $T_1 = 200^\circ\text{C}$ and $T_2 = 0^\circ\text{C}$ its efficiency is η_1 and when temperature $T_1 = 0^\circ\text{C}$ and $T_2 = -200^\circ\text{C}$ its efficiency is η_2 then find the value of $\frac{\eta_1}{\eta_2} = ?$
- (1) $\frac{273}{473}$ (2) $\frac{473}{273}$ (3) $\frac{373}{273}$ (4) $\frac{273}{373}$

44. The temperature of the two outer surfaces of a composite slab, consisting of two materials having coefficients of thermal conductivity K and 2K and thickness x and 4x, respectively are T_2 and T_1 ($T_2 > T_1$). The rate of heat transfer through the slab, in a steady state is $\left(\frac{A(T_2 - T_1)K}{x} \right) f$, with f is equal to :-



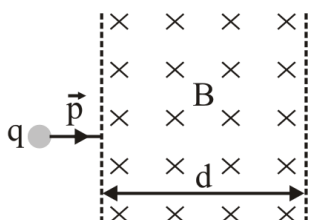
- (1) 1 (2) $\frac{1}{2}$ (3) $\frac{2}{3}$ (4) $\frac{1}{3}$
45. Two metal rods, 1 & 2 of same length have same temperature difference between their ends, their thermal conductivities are K_1 & K_2 and cross sectional areas A_1 & A_2 respectively. What is required condition for same rate of heat conduction in them.

- (1) $K_1 = K_2$ (2) $K_1 A_1 = K_2 A_2$
 (3) $\frac{K_1}{A_1} = \frac{K_2}{A_2}$ (4) $\frac{K_1}{\ell_1^2} = \frac{K_2}{\ell_2^2}$

46. The magnetic force acting on a charge particle of charge $-2\mu\text{C}$ in a magnetic field of 2T acting in y direction, when the particle velocity is $(2\hat{i} + 3\hat{j}) \times 10^6 \text{ ms}^{-1}$ is

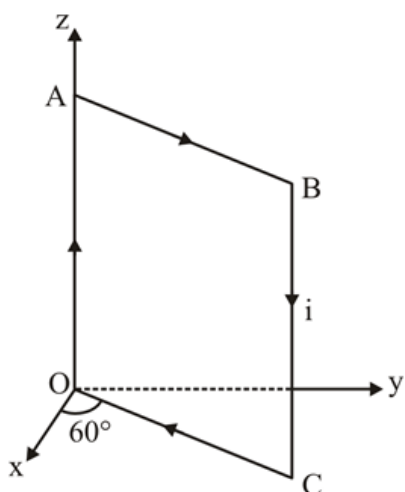
- (1) 8N in $-z$ direction (2) 4N in z direction
(3) 8N in y direction (4) 8N in z direction

47. A particle with charge q , moving with a momentum p , enters a uniform magnetic field normally. The magnetic field has magnitude B and is confined to a region of width d , where $d < \frac{p}{Bq}$. The particle is deflected by an angle θ in crossing the field.



- (1) $\sin\theta = \frac{Bqd}{p}$ (2) $\sin\theta = \frac{p}{Bqd}$
(3) $\sin\theta = \frac{Bp}{qd}$ (4) $\sin\theta = \frac{pd}{Bq}$

48. A square loop OABCO of side L carries a current i . It is placed as shown in figure. The magnetic moment of the loop will be :



- (1) $\frac{iL^2}{2}(\sqrt{3}\hat{i} - \hat{j})$ (2) $\frac{iL^2}{2}(\hat{i} - \hat{j}\sqrt{3})$
(3) $\frac{iL^2}{2}(-\sqrt{3}\hat{i} + \hat{j})$ (4) $\frac{iL^2}{2}(-\hat{i} + \sqrt{3}\hat{j})$

49. If a charge is projected in a zero gravity region then match the condition in column I with possible cases of electric field and magnetic field given in column II

	Column I		Column II
A.	No net force exerts on charge	P.	$E = 0, B = 0$
B.	Charge moves without change in its velocity	Q.	$E = 0, B \neq 0$
C.	Charge moves without deviation or deflection	R.	$E \neq 0, B \neq 0$
D.	Charge moves without change in its speed	S.	$E \neq 0, B = 0$

- (1) $A \rightarrow PQR; B \rightarrow PQRS; C \rightarrow PQRS; D \rightarrow PQR$
(2) $A \rightarrow PQRS; B \rightarrow PQR; C \rightarrow PQRS; D \rightarrow PQR$
(3) $A \rightarrow PQR; B \rightarrow PQR; C \rightarrow PQRS; D \rightarrow PQR$
(4) $A \rightarrow PQRS; B \rightarrow PQRS; C \rightarrow PQR; D \rightarrow PQR$
50. When a proton is released from rest in a room, it starts with an initial acceleration a_0 towards the east. When it is projected towards the north with a speed v_0 , it moves with initial acceleration $3a_0$ towards east. The electric and magnetic fields in the room are :
- (1) $\frac{Ma_0}{e}$ west, $\frac{Ma_0}{ev_0}$ up
(2) $\frac{Ma_0}{e}$ west, $\frac{2Ma_0}{ev_0}$ down
(3) $\frac{Ma_0}{e}$ east, $\frac{2Ma_0}{ev_0}$ down
(4) $\frac{Ma_0}{e}$ east, $\frac{2Ma_0}{ev_0}$ up

Topic : OXIDATION, REDUCTION, NAME REACTION, POC, CHEMICAL BONDING UPTO DIPOLE MOMENT, CHEMICAL EQUILIBRIUM [BACK UNIT] : NOMENCLATURE, ISOMERISM

SECTION-A (CHEMISTRY)

51. The species having bond angles of 120° is ____.

- (1) PH_3 (2) ClF_3
(3) NCl_3 (4) BCl_3

52. In $\text{C}_3\text{O}_2(\text{O}=\text{C}=\text{C}=\text{O})$, the hybridization state of C is -

- (1) sp^2 (2) sp
(3) sp^3 (4) sp^3d^2

53. Variable covalency is exhibited by

- (1) P and S
(2) N and O
(3) N and P
(4) F and Cl

54. Which of the following combination of atomic orbitals give probability of covalent bond formation ?

- (1) $p_x + p_y$
(2) $d_{xy} + d_{yz}$
(3) $p_z + d_{xy}$
(4) $s + p_x$

55. Maximum number of lone pair of electrons on central atom are present in:

- (1) ClO_3^-
(2) XeF_4
(3) SF_4
(4) I_3^-

56. Which of the following is odd electron molecule ?

- (1) NO (2) CH_4
(3) Cl_2O_7 (4) N_2O_5

57. Which of the following order of Bond angle is correct ?

- (1) $\text{H}_2\text{O} > \text{NH}_3 > \text{CH}_4$
(2) $\text{CH}_4 > \text{NH}_3 > \text{H}_2\text{O}$
(3) $\text{H}_2\text{O} > \text{CH}_4 > \text{NH}_3$
(4) $\text{NH}_3 > \text{H}_2\text{O} > \text{CH}_4$

58. Which of the following molecule is more polar than other ?

- (1) NH_3 (2) NF_3 (3) BF_3 (4) CO_2

59. Which of the following are isostructural ?

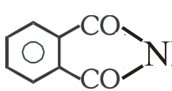
- (1) XeF_2 & CO_2 (2) N_3^- & BeF_2
(3) N_2O & BeCl_2 (4) All

60. Which of the following having π -bonds only ?

- (1) B_2 (2) C_2 (3) O_2 (4) 1 & 2

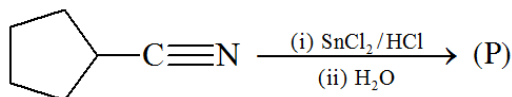
61. Which of the following is correct sequence of energy of molecular orbital in N_2 molecule ?

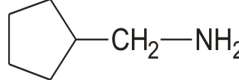
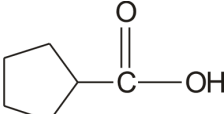
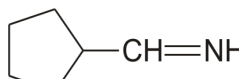
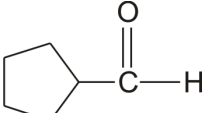
- (1) $\sigma 1s^2 < \sigma^* 1s^2 < \sigma 2s^2 < \sigma^* 2s^2 < \sigma 2p_z^2 < \pi 2p_x^2 = \pi 2p_y^2 < \sigma^* 2p_z^0 < \pi^* 2p_x^0 = \pi^* 2p_y^0$
(2) $\sigma 1s^2 > \sigma^* 1s^2 < \sigma 2s^2 < \sigma^* 2s^2 < \pi 2p_x^2 = \pi 2p_y^2 < \sigma 2p_z^0 < \pi^* 2p_x^1 = \pi^* 2p_y^1 < \sigma^* 2p_z^0$
(3) $\sigma 1s^2 < \sigma^* 1s^2 < \sigma 2s^2 < \sigma^* 2s^2 < \pi 2p_x^2 = \pi 2p_y^2 < \sigma 2p_z^2 < \pi^* 2p_x^0 = \pi^* 2p_y^0 < \sigma^* 2p_z^2$
(4) None of these

62.  $\xrightarrow[\text{(ii) C}_2\text{H}_5\text{Br}]{\text{(i) KOH}}$ A $\xrightarrow{\text{H}_3\text{O}^+}$ B + C
B and C in the above reaction sequence are

- (1) Benzoic acid + aniline
(2) Phthalic acid + ethylamine
(3) Phthalic acid + aniline
(4) Benzoic acid + ethylamine

63. The product (P) in the following reaction is :

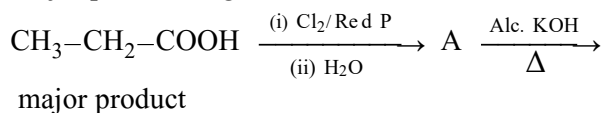


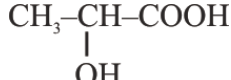
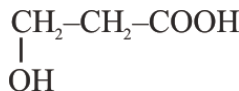
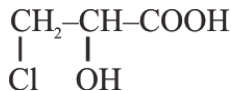
- (1)  (2) 
 (3)  (4) 

64. $\text{R}-\text{C}(=\text{O})-\text{NH}_2 \xrightarrow[\text{(2) H}_3\text{O}^+]{\text{(1) LiAlH}_4} \text{P (Major)}$
 P should be :-

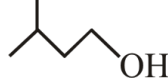
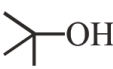
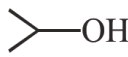
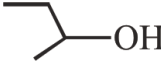
- (1) $\text{R}-\text{CH}_2-\text{OH}$ (2) $\text{R}-\text{CH}_2-\text{NH}_2$
 (3) $\text{R}-\text{C}(=\text{O})-\text{H}$ (4) None of these

65. Major product of given reaction :-

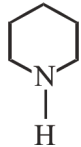
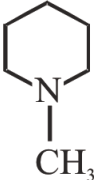


- (1) 
 (2) 
 (3) $\text{CH}_2=\text{CH}-\text{COOH}$
 (4) 

66. In Victor Meyer's test, red colour is given by—

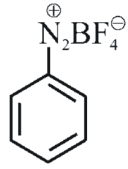
- (1)  (2) 
 (3)  (4) 

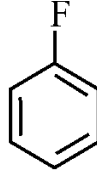
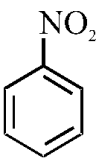
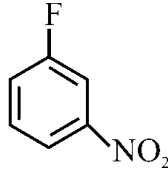
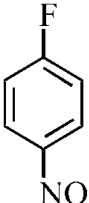
67. Which of the following give positive isocyanide test :-

- (1)  (2) CH_3-NH_2
 (3) $\text{CH}_3-\text{NH}-\text{CH}_3$ (4) 

68. Which of the following compound does not liberate CO_2 gas on reaction with NaHCO_3 ?

- (1) Benzoic acid
 (2) Picric Acid
 (3) Phenol
 (4) Benzene Sulphonic Acid

69.  $\xrightarrow[\text{Cu}/\Delta]{\text{NaNO}_2}$ Product
 Product of above reaction is :-

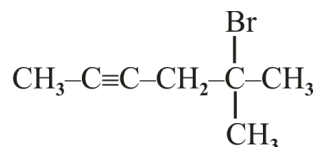
- (1)  (2) 
 (3)  (4) 

70. $\text{CH}_3-\text{CH}_2-\text{NH}-\text{CHO}$ (I) $\text{CH}_3-\text{CH}(\text{NH}_2)-\text{CHO}$ (II)

Which type of isomerism is observed between I and II :-

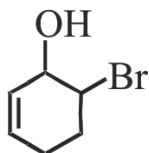
- (1) Chain isomerism
 (2) Position isomerism
 (3) Functional isomerism
 (4) Metamerism

71. What is the IUPAC name for the following alkyne ?



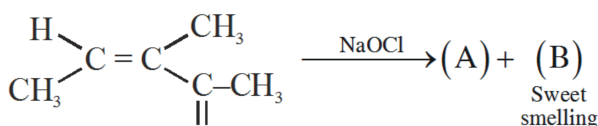
- (1) 5-Bromo-2-heptyne
 (2) 3-Bromo-5-heptyne
 (3) 2-Bromo-2-methyl-4-hexyne
 (4) 5-Bromo-5-methyl-2-hexyne

72. The correct IUPAC name of :-



- (1) 2-Bromo cyclo hex-5-enol
- (2) 1-Bromo-2-hydroxy cyclohex-3-ene
- (3) 3-Hydroxy-4-bromo cyclohexene
- (4) 6-Bromo cyclohex-2-enol

73.



Identify (A) & (B) respectively :-

- (1) + CHCl_3
- (2) + $\text{H}-\text{C}(=\text{O})-\text{CH}_3$
- (3) + $\text{Cl}-\text{C}(=\text{O})-\text{Cl}$
- (4) + CHCl_3

74. Ethanol reacts with H_2SO_4 at 413 K to form :-

- (1) Ethene
- (2) Ethyl hydrogen sulphate
- (3) Ethoxy ethane
- (4) None of these

75. Carboxylic acid can be reduced into primary alcohol by :

- (1) By LiAlH_4
- (2) $\text{P} + \text{HI}$
- (3) By NaBH_4
- (4) (1) & (2) both

76. PCl_5 dissociated in a closed container as :-



If total pressure at equilibrium of the reaction mixture is P and degree of dissociation of PCl_5 is α , the partial pressure of PCl_3 will be :-

- (1) $P \cdot \left[\frac{\alpha}{\alpha + 1} \right]$
- (2) $P \cdot \left[\frac{2\alpha}{1 + \alpha} \right]$
- (3) $P \cdot \left[\frac{\alpha}{\alpha - 1} \right]$
- (4) $P \cdot \left[\frac{\alpha}{1 - \alpha} \right]$

77. For the reaction $2\text{CO}_2 \rightleftharpoons 2\text{CO} + \text{O}_2$, the dissociation of 2 moles of CO_2 is carried out in container with volume 5L, if at equilibrium 60% CO_2 is dissociated, then total number of moles at equilibrium will be :-

- (1) 2.3 (2) 2.6 (3) 0.52 (4) 3.2

78. When pressure is applied to the equilibrium system $\text{H}_2\text{O(s)} \rightleftharpoons \text{H}_2\text{O(l)}$ which of the following phenomenon will take place :-

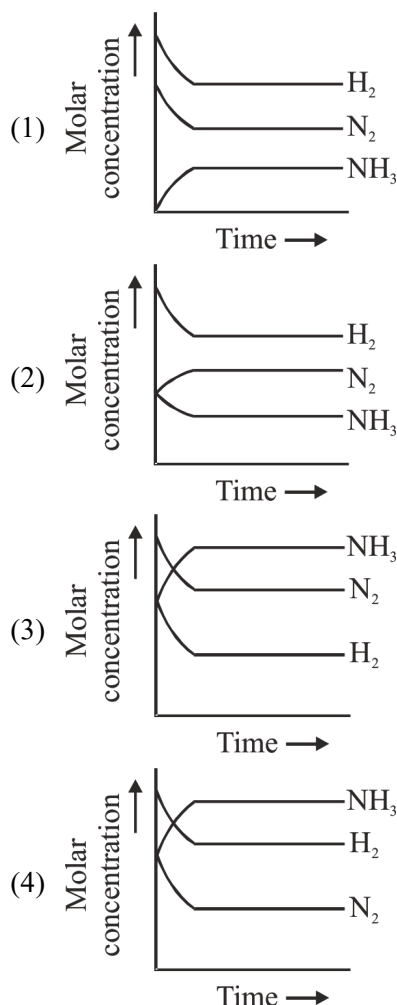
- (1) More ice will be formed
- (2) Ice will sublime
- (3) More water will be formed
- (4) Equilibrium will not disturb

79. For the reaction $\text{PCl}_5(g) \rightleftharpoons \text{PCl}_3(g) + \text{Cl}_2(g)$, the forward reaction at constant temperature is favoured by :-

- (i) Introducing inert gas at constant volume
- (ii) Introducing Cl_2 gas at constant volume
- (iii) Introducing an inert gas at constant pressure
- (iv) Increasing the pressure of the system
- (v) Introducing PCl_5

- (1) (iii), (iv) & (v)
- (2) (iii) & (v)
- (3) (i), (ii) & (v)
- (4) Only (iii)

80. For the synthesis of ammonia by the reaction $\text{N}_2 + 3\text{H}_2 \rightleftharpoons 2\text{NH}_3$ in the Haber's process, the attainment of equilibrium is correctly predicted by the curve :-



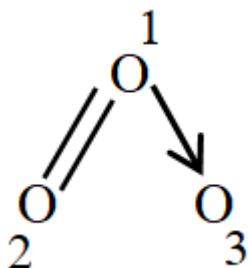
81. For which of the following reaction, increase in pressure favours the forward reaction?
- $[\text{Ni}(\text{CO})_4](\text{g}) \rightleftharpoons \text{Ni}(\text{s}) + 4\text{CO}(\text{g})$
 - $\text{PCl}_5(\text{g}) \rightleftharpoons \text{PCl}_3(\text{g}) + \text{Cl}_2(\text{g})$
 - $\text{H}_2(\text{g}) + \text{I}_2(\text{g}) \rightleftharpoons 2\text{HI}(\text{g})$
 - $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$
82. For the reaction $\text{N}_2\text{O}_4(\text{g}) \rightleftharpoons 2\text{NO}_2(\text{g})$, if initially only 4 atm N_2O_4 is present and at equilibrium, total pressure of equilibrium mixture is 6 atm then the value of K_p is
- 2 atm
 - 4 atm
 - 8 atm
 - 16 atm

83. If value of K_c for an equilibrium is 2.4×10^{47} then which of the following statement is correct?
- Reactants predominate over products
 - Products predominate over reactants
 - Reaction proceeds rarely
 - Both the reactants and products are present in appreciable amount
84. Unit of K_c for the equilibrium $\text{CaCO}_3(\text{s}) \rightleftharpoons \text{CaO}(\text{s}) + \text{CO}_2(\text{g})$ is
- mol L^{-1}
 - $\text{mol}^2 \text{L}^{-2}$
 - L mol^{-1}
 - $\text{L}^2 \text{mol}^{-2}$
85. If equilibrium constant for the reaction $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$ is K_c then the equilibrium constant for the reaction $\text{NH}_3(\text{g}) \rightleftharpoons \frac{1}{2} \text{N}_2(\text{g}) + \frac{3}{2} \text{H}_2(\text{g})$ is
- K_c
 - $\frac{1}{K_c}$
 - $\frac{1}{\sqrt{K_c}}$
 - $\sqrt{K_c}$

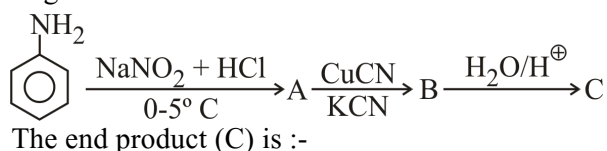
SECTION-B (CHEMISTRY)

86. In which of the following process hybridisation of central atom is changed : –
- $\text{H}_2\text{O} + \text{H}^+ \rightarrow \text{H}_3\text{O}^+$
 - $\text{NH}_3 + \text{H}^+ \rightarrow \text{NH}_4^+$
 - $\text{BF}_3 + \text{F}^- \rightarrow \text{BF}_4^-$
 - $2\text{NO}_2 \rightarrow \text{N}_2\text{O}_4$
87. In which of the following species $p\pi$ - $d\pi$ bond is present but $p\pi$ - $p\pi$ bond is absent?
- SiH_4
 - CS_2
 - SO_2
 - SO_2Cl_2
88. Correct order of dipole moment is :
- $\text{BF}_3 < \text{H}_2\text{S} < \text{H}_2\text{O}$
 - $\text{H}_2\text{S} < \text{BF}_3 < \text{H}_2\text{O}$
 - $\text{H}_2\text{O} < \text{H}_2\text{S} < \text{BF}_3$
 - $\text{BF}_3 < \text{H}_2\text{O} < \text{H}_2\text{S}$

89. What is the formal charge on ^1O , ^2O & ^3O respectively in given molecule.

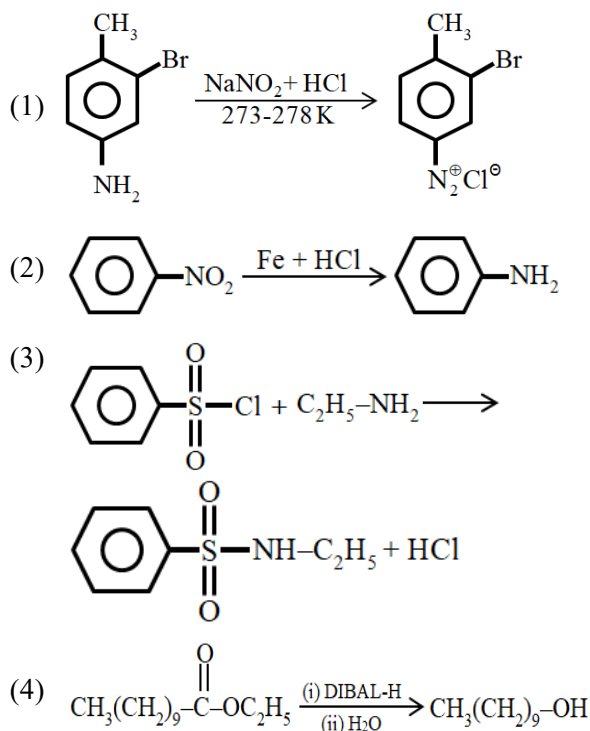


- (1) +1, -1, 0
 (2) +1, 0, -1
 (3) 0, +1, -1
 (4) -1, +1, 0
90. Which of the following overlapping of orbitals results in delta bond formation ?
- (1) $d_{xy} + d_{xy}$ at x axis
 (2) $d_{yz} + p_x$ at z axis
 (3) $d_{zx} + d_{zx}$ at y axis
 (4) $s + p_x$ at x axis
91. In given reaction

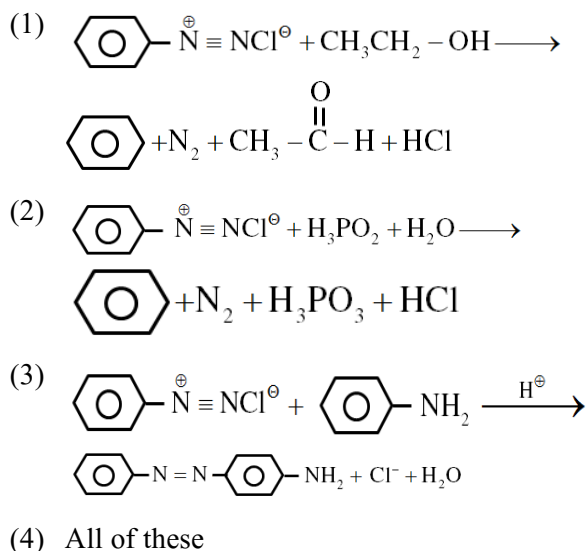


- (1)
- (2)
- (3)
- (4)

92. Which of the following is incorrect?



93. Which of the following reaction is an example of electrophilic substitution reaction ?



94. Which of the following statement is incorrect regarding gabriel phthalimide synthesis.

- (1) It is used for the preparation of primary aliphatic amines.
 (2) Aromatic primary amines cannot be prepared by this method.
 (3) Aryl halides do not undergo nucleophilic substitution reaction with the anion formed by phthalimide
 (4) Aniline can be prepared using this method.

95. Match the column.

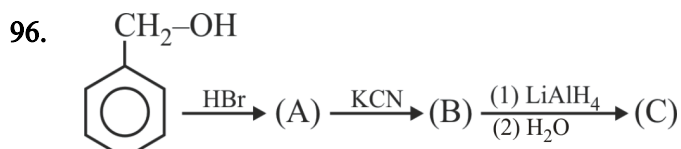
	Column-I		Column-II
(A)		(P)	Resorcinol
(B)		(Q)	Catechol
(C)		(R)	Quinol
(D)		(S)	m-cresol

(1) (A)-Q, (B)-R, (C)-P, (D)-S

(2) (A)-R, (B)-Q, (C)-P, (D)-S

(3) (A)-P, (B)-R, (C)-Q, (D)-S

(4) (A)-S, (B)-P, (C)-Q, (D)-R



What is A, B & C respectively ?

- (1) ,
- (2) ,
- (3) ,
- (4) ,

97.

The equilibrium constant (K_p) for the reaction $\text{H}_2\text{O}(\text{g}) \rightleftharpoons \text{H}_2(\text{g}) + \frac{1}{2}\text{O}_2(\text{g})$ is related to degree of dissociation (α) at a total pressure P, given by :

$$(1) K_p = \frac{\alpha^3 p^{1/2}}{(1 + \alpha)(2 + \alpha)^{1/2}}$$

$$(2) K_p = \frac{\alpha^3 p^{3/2}}{(1 - \alpha)(2 + \alpha)^{1/2}}$$

$$(3) K_p = \frac{\alpha^{3/2} p^2}{(1 - \alpha)(2 + \alpha)^{1/2}}$$

$$(4) K_p = \frac{\alpha^{3/2} p^{1/2}}{(1 - \alpha)(2 + \alpha)^{1/2}}$$

98.

The equilibrium $\text{P}_4(\text{g}) + 6\text{Cl}_2(\text{g}) \rightleftharpoons 4\text{PCl}_3(\text{g})$ is attained by mixing equal moles of P_4 and Cl_2 in an evacuated vessel. Then at equilibrium always true is :-

(1) $[\text{Cl}_2] > [\text{PCl}_3]$

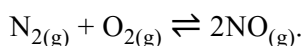
(2) $[\text{Cl}_2] > [\text{P}_4]$

(3) $[\text{P}_4] > [\text{Cl}_2]$

(4) $[\text{PCl}_3] > [\text{P}_4]$

99.

At a certain temperature equilibrium constant $K_C = 4 \times 10^{-2}$ for the reaction,



If we take 1.5 mole of NO & 2 mole each of N_2 & O_2 in 5L vessel, what would be the equilibrium concentration of NO (in mol L^{-1}) :

(1) 0.1 M

(2) 0.5 M

(3) 0.8 M

(4) 1 M

100.

Select the correct relation :

(1) $\Delta G = \Delta G^\circ + RT \log K$

(2) $\Delta G^\circ = \Delta G + RT \ln Q$

(3) $\Delta G = \Delta G^\circ + RT \ln K$

(4) $\Delta G^\circ = \Delta G + RT \ln K$

Topic : LAC OPERON TO MENDELISM, MONERA (NUTRITION, REPRODUCTION, ECO. IMP.) BGA, MYCOPLASMA, ARCHAE BAC.) KINGDOM-PROTISTA, FUNGI [BACK UNIT] : ECOLOGY

SECTION-A (BIOLOGY-I)

- 101.** Which of the following smallest organism completely lack cell wall, pathogenic in animals and plants and can survive with oxygen ?
 (1) Archaeobacteria (2) Eubacteria
 (3) Cyanobacteria (4) Mycoplasma
- 102.** With respect to sexual reproduction in fungi the correct sequence of events is
 (1) Karyogamy, Plasmogamy and Meiosis
 (2) Meiosis, Plasmogamy and Karyogamy
 (3) Plasmogamy, Karyogamy and Meiosis
 (4) Meiosis, Karyogamy and Plasmogamy
- 103.** Fungal spores produced asexually and exogenously at the tip of hyphae are
 (1) Conidia (2) Zoospore
 (3) Sporangiospore (4) All of these
- 104.** Which of the following organisms are known as chief producers in the oceans ?
 (1) Cyanobacteria (2) Diatoms
 (3) Dinoflagellates (4) Euglenoids
- 105.** Which is commonly called "*Drosophilla* of plant kingdom" :-
 (1) *Morchella* (2) *Neurospora*
 (3) *Rhizopus* (4) *Claviceps*
- 106.** Euglenoids have
 (1) Two flagella of same size
 (2) Lipid rich layer called pellicle
 (3) Pigments identical to those present in higher plants
 (4) Only heterotrophic mode of nutrition even in the presence of sunlight
- 107.** In which of the following plasmogamy is followed by karyogamy, immediately?
 (1) Mucor (2) Aspergillus
 (3) Ustilago (4) Puccinia
- 108.** Some of Cyanobacteria fix nitrogen in a specialised cell, which is
 (1) Thin walled
 (2) Green cell
 (3) Called hormogonia
 (4) Thick walled and non green
- 109.** What is common in Ustilago, Neurospora and Agaricus ?
 (1) All are edible
 (2) All have a dikaryotic phase in their life cycle
 (3) All are commonly called 'sac fungi'
 (4) All produce motile sexual spores
- 110.** A free living nitrogen - fixing cyano-bacterium which can also form symbiotic association with the water fern *Azolla* is -
 (1) *Anabaena* (2) *Oscillatoria*
 (3) *Chlorella* (4) *Nostoc*
- 111.** Match the column correctly -

Column I		Column II	
A	T.B.	i	<i>Mycobacterium</i>
B	Citrus canker	ii	<i>Xanthomonas</i>
C	Tetanus	iii	<i>Clostridium</i>
D	Cholera	iv	<i>Vibrio cholera</i>

- (1) A - (i), B - (ii), C - (iii), D - (iv)
 (2) A - (ii), B - (i), C - (iii), D - (iv)
 (3) A - (iii), B - (ii), C - (i), D - (iv)
 (4) A - (iv), B - (iii), C - (ii), D - (i)

112. Read the following given words.

Dolipore, Clamp connection, Smut disease, Tuffels, Damping off disease in vegetable
How many of them related to Basidiomycetes ?

- (1) 2 (2) 3 (3) 4 (4) 5

113. Identify the figure and product given below:-

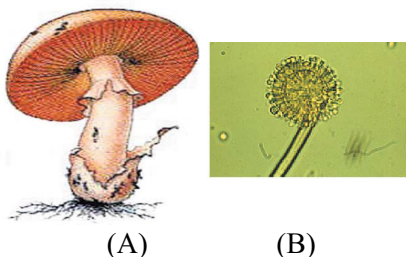


	Figure-A	Figure-B
(1)	<i>Aspergillus</i>	<i>Puccinia</i>
(2)	<i>Puccinia</i>	<i>Agaricus</i>
(3)	<i>Aspergillus</i>	<i>Neurospora</i>
(4)	<i>Agaricus</i>	<i>Aspergillus</i>

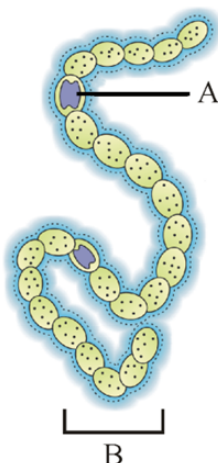
114. Which of the following bacteria uses H_2S as e^- and H^+ donor for its nutrition.

- (1) *Chromatium* (2) *Thiobacillus*
(3) *Rhodospirillum* (4) *Nitrosomonas*

115. _____ are present in gut of several ruminants.

- (1) Halophiles (2) BGA
(3) Methanogens (4) Thermoacidophiles

116. Identify A and B in the given diagram



- (1) A-Heterocyst
B-Filamentous BGA

- (2) A-Heterocyst
B-Colonial BGA

- (3) A-Trichoderm cell
B-Nostoc

- (4) A-Mucilagenous sheath
B-Spirulina

117. The group of diatoms and desmids is

- (1) Chrysophyta
(2) Pyrrophyta
(3) Cyanophyta
(4) Euglenoids

118. Diatom do not show

- (1) Photosynthetic mode of nutrition
(2) Two thick overlapping shell
(3) Binary fission
(4) Large amount of cell wall deposits in their habitat.

119. All are the criteria for classification of fungi, except

- (1) Composition of cell wall
(2) Morphology of the mycelium
(3) Mode of spore formation
(4) Fruiting bodies

120. In imperfect fungi, when the sexual forms are discovered they are often moved to

- (1) Ascomycetes
(2) Basidiomycetes
(3) Both (1) & (2)
(4) Deuteromycetes

121. Identify the fungi on the basis of statements

A and B choose the correct option:

A - Extensively used in genetic work

B - Used for making bread and beer

- (1) A - *Neurospora* (2) A - *Penicillium*
 B - *Penicillium* B - Yeast
 (3) A - *Neurospora* (4) A - *Trichoderma*
 B - Yeast B - *Aspergillus*

122. Absence of cell wall, mixotrophic mode of nutrition, locomotion by flagella and longitudinal binary fission. All these features are shown by

- (1) Diatoms
 (2) Euglenoid
 (3) Dinoflagellates
 (4) Slime mould

123. Mycoplasma are insensitive to ____ (a) ____ and sensitive to ____ (b) ____ antibiotic

	(a)	(b)
(1)	Streptomycin	Penicillin
(2)	Erythromycin	Streptomycin
(3)	Chloramphenicol	Penicillin
(4)	Penicillin	Chloramphenicol

124. Find the incorrectly matched pair w.r.t. fungi and their sexual spore :

- (1) *Albugo* - Oospore
 (2) *Penicillium* - Ascospore
 (3) *Ustilago* - Basidiospore
 (4) *Alternaria* - Zygosporangium

125. Endogenous haploid sexual spores are characteristic of class

- (1) Ascomycetes
 (2) Phycomycetes
 (3) Basidiomycetes
 (4) Deuteromycetes

126. Given below are two statements :

Statement-I : Troposphere is the zone of atmosphere in which the good Ozone is present.

Statement-II : Montreal protocol was signed in 1987 for the control of emission of Ozone depleting substances.

- (1) Both Statement-I and Statement-II are incorrect
 (2) Statement-I is correct but Statement-II incorrect
 (3) Statement-I is incorrect but Statement-II correct
 (4) Both Statement-I and Statement-II are correct

127. World-summit on sustainable development held in 2002 in ____ :-

- (1) Johannesburg (2) Kyoto
 (3) Rio de Janeiro (4) Copenhagen

128. Which of the following pollutants have maximum Ozone depleting potential ?

- (1) Nitrogen oxide (2) CFC
 (3) CH₄ (4) CO

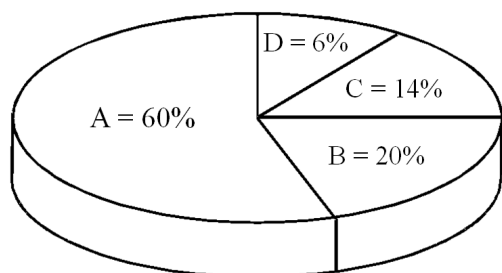
129. The minimum DDT concentration in aquatic food chain shall occur in :-

- (1) Primary consumer
 (2) Primary producer
 (3) Secondary producer
 (4) Top consumer

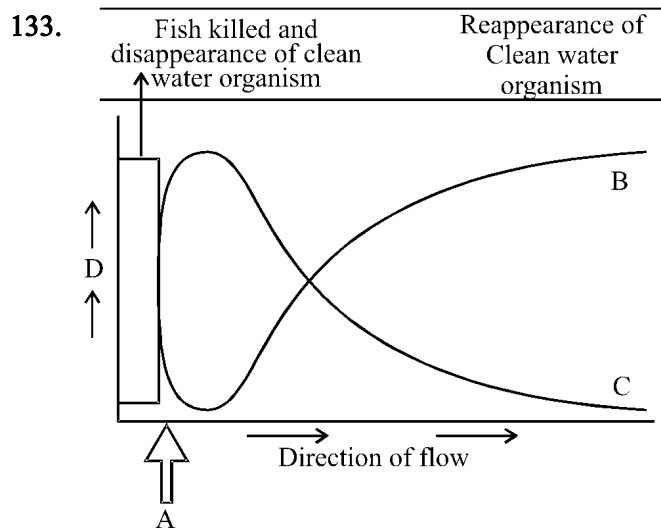
130. Which metal used as catalytic converter in vehicles.

- (1) Platinum
 (2) Palladium
 (3) Rhodium
 (4) All of the above

131. In the given pie diagram of greenhouse gases, A, B, C and D are represented by :-



- (1) A = CO₂, B = CFC, C = CH₄, D = N₂O
 (2) A = CO₂, B = N₂O, C = CH₄, D = CFC
 (3) A = CH₄, B = CO₂, C = CFC, D = N₂O
 (4) A = CO₂, B = CH₄, C = CFC, D = N₂O
132. According to Central Pollution Control Board (CPCB) which size of particulate are responsible for causing the greatest harm to human health :-
- (1) 2.5 micrometer or less than 2.5 micrometer
 (2) 3.5 micrometer or less than 3.5 micrometer
 (3) 4.5 micrometer or less than 4.5 micrometer
 (4) 5.5 micrometer or less than 5.5 micrometer



- (1) A–Sewage discharge, B–BOD, C–DO, D–Concentration
 (2) A–Sewage discharge, B–DO, C–BOD, D–Concentration
 (3) A–Concentration, B–BOD, C–DO, D–Sewage discharge
 (4) A–Concentration, B–DO, C–BOD, D–Sewage discharge

134. Match the items in column I and column II and choose the correct option :-

Column I		Column II	
A	UV	i	Biomagnification
B	Biodegradable organic matter	ii	Eutrophication
C	DDT	iii	Snow blindness
D	Phosphates	iv	BOD

The correct match is :-

- (1) A-ii, B-i, C-iv, D-iii (2) A-iii, B-ii, C-iv, D-i
 (3) A-iii, B-iv, C-i, D-ii (4) A-iii, B-i, C-iv, D-i
135. How many statements are correct :-
- (a) Eutrophication is natural aging of a lake by nutrient enrichment of its water
 (b) After CO₂, methane (CH₄) is major cause of green house effect
 (c) Ozone is secondary pollutant in troposphere
 (d) Thickness of ozone is measured in dobson unit.
- (1) a,b (2) a,b,c (3) a,b,c,d (4) b,c,d

SECTION-B (BIOLOGY-I)

136. Match the column and select the correct option for the following.

	Column-I		Column-II
1.	Phycomycetes	(i)	Diatomaceous earth
2.	Dinoflagellates	(ii)	Puccinia
3.	Basidiomycetes	(iii)	Aseptate, coenocytic
4.	Chrysophytes	(iv)	Gonyaulax

- (1) 1 – iii, 2 – iv, 3 – ii, 4 – i
 (2) 1 – ii, 2 – iv, 3 – iii, 4 – i
 (3) 1 – iii, 2 – i, 3 – ii, 4 – iv
 (4) 1 – iii, 2 – ii, 3 – i, 4 – iv
137. Dinoflagellates are
- (1) Mostly marine
 (2) Mostly freshwater forms
 (3) Mostly symbiotic
 (4) Mostly found at moist and damp places.

138. Consider the following statements :-

- (a) they grow in soil, on logs and tree stumps and in living plant bodies as parasites.
- (b) The mycelium is branched and septate and the asexual spores are generally not found but vegetative reproduction by fragmentation is common.
- (c) The sex organs are absent but plasmogamy is brought about by fusion of two vegetative or somatic cells of different strains or genotypes.

Above statements are related to the member of which class :-

- (1) Phycomycetes (2) Ascomycetes
- (3) Basidiomycetes (4) Deuteromycetes

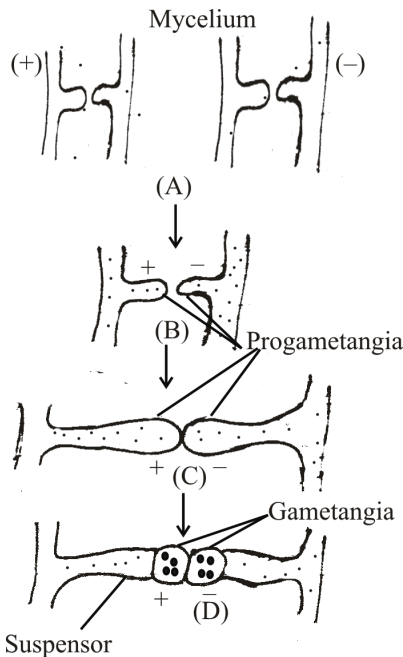
139. List some of the fungi is given below :-

Alternaria, Trichoderma, Ustilago, Mucor, Rhizopus, Albigo

How many are members of class deuteromycetes?

- (1) Three (2) Two (3) Four (4) Five

140.



Suspensor

The figures represent stages in the sexual reproduction in Rhizopus. If gametangia are fused, then what is the fate of cells ?

- (1) They will develop into a zoospore
- (2) They will develop into a prothallus
- (3) They will develop into a parthenospore.
- (4) They will develop into a zygospore

141. Choose the correct option with reference to Photosynthetic protista –

Group		Flagella on main body	Nutrition	Habitat
(1)	Diatoms	Present	Phototrophs	Mainly marine
(2)	Dinoflagellate	Present	Phototrophs	Mainly marine
(3)	Euglenoids	Absent	Phototrophs & Heterotrophs	Mainly fresh water
(4)	Slime moulds	Absent	Autotrophs	Decaying twing and leaves

142. Fungi show :

- (A) Great diversity in morphology and Habitat
 - (B) Abundance in warm and humid place
 - (C) Cell wall made up of cellulose and polysaccharides.
- The correct option is :-

- (1) Only A (2) Only A and B
- (3) Only C (4) All correct

143. Diatomaceous earth is :

- (1) Destructible cell envelope of smallest living cell
- (2) Indestructible Silicious cell wall of fire algae
- (3) Indestructible CaCO_3 shells of amoeboid protozoans.
- (4) Indestructible Silicious cell wall of phytoplankton

144. Read the following features carefully and identify the class of fungi.

- (i) Asepted and coenocytic mycelium
 - (ii) The asexual reproduction by spore
 - (iii) Members are found in aquatic habitats and on decaying wood in moist and damp places or as obligate parasites on plants
- (1) Ascomycetes
 - (2) Deuteromycetes
 - (3) Phycomycetes
 - (4) Basidiomycetes.

145. Choose the correct set of fungi that exhibit diakaryophase in life cycle.

- (1) *Puccinia* and *Albugo*
- (2) *Puccinia* and *Claviceps*
- (3) *Albugo* and *Aspergillus*
- (4) *Agaricus* and *Rhizopus*

146. Match the following column correctly :-

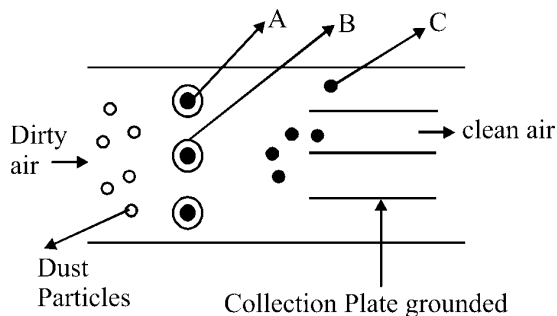
	Column-I		Column-II
A.	Photochemical smog	(i)	Coal burning
B.	Montreal protocol	(ii)	Climate change
C.	Sulphur smog	(iii)	PAN
D.	Kyoto protocol	(iv)	Ozone layer depletion

- (1) A–(iii), B–(iv), C–(ii), D–(i)
- (2) A–(ii), B–(iv), C–(iii), D–(i)
- (3) A–(i), B–(ii), C–(iii), D–(iv)
- (4) A–(iii), B–(iv), C–(i), D–(ii)

147. Green Mufflar Scheme is related to controlling of :-

- (1) Pesticide pollution
- (2) Sound pollution
- (3) Air pollution
- (4) Water pollution

148.



Choose the correct about above diagram

TG: @Chalnaayaaar

	Name of device	Principle	B	C
(1)	Wet scrubber	Gravity	Lime particles	Dust particle
(2)	Cyclon	Water sedimentation	Water particles	+ve charged particles
(3)	E.S.P.	Electric charging	Corona	–ve charged particles
(4)	E.S.P.	Electric charging	–ve charged wire	+ve charged particles

149. Match the column I to II.

	Column-I		Column-II
(a)	Catalytic converter	(i)	Particulate matter
(b)	Electrostatic precipitator	(ii)	CO and NO _x
(c)	Earmuffs	(iii)	High noise pollution
(d)	Land fills	(iv)	Solid wastes

- (1) a – ii, b – i, c – iii, d – iv
- (2) a – i, b – ii, c – iii, d – iv
- (3) a – iv, b – iii, c – ii, d – i
- (4) a – ii, b – iii, c – iv, d – i

150. Which of the statement is/are correct?

- (a) Catalytic converters have expensive metals like platinum, palladium and silver as catalyst
 - (b) Air act was amended in year 1987 to include noise as a air pollutant
 - (c) A scrubber can remove gases like sulphur dioxide
 - (d) ESP can remove 99% particulate matter from exhaust of vehicles.
- (1) a & b are correct (2) b & c are correct
 - (3) c & d are correct (4) d & a are correct

Topic : LAC OPERON TO MENDELISM, MONERA (NUTRITION, REPRODUCTION, ECO. IMP.) BGA, MYCOPLASMA, ARCHAE BAC.) KINGDOM-PROTISTA, FUNGI [BACK UNIT] : ECOLOGY

SECTION-A (BIOLOGY-II)

- 151. Statement-1 :** Eastern Ghats have a greater amphibian species diversity than the Western Ghats.
- Statement-2 :** Species diversity increase as we move away from the equator towards the poles.
- (1) Both Statement I and Statement II are incorrect.
- (2) Statement I is correct but Statement II is incorrect.
- (3) Statement I is incorrect but Statement II is correct
- (4) Both Statement I and Statement II are correct.
- 152.** Relationship between yucca plant flowers and Pronuba insects:
- (1) Commensalism (2) Mutualism
- (3) Amensalism (4) Parasitism
- 153.** Which is not essential for self sustaining ecosystem?
- (1) Producers
- (2) Macro consumers
- (3) Micro consumers
- (4) Decomposer
- 154.** The principle of competitive release was stated by :
- (1) Mac Arthur (2) G.F. Gause
- (3) Conell (4) Darwin
- 155.** Which one of the following interaction is widely used in medical science for the production of antibiotics?
- (1) Commensalism
- (2) Mutualism
- (3) Parasitism
- (4) Amensalism
- 156.** Which one of the following pair of organism produce a large number of small sized offspring?
- (1) Pacific salmon fish, Bamboo
- (2) Oysters, Pelagic fish
- (3) Oysters, Pacific salmon fish
- (4) Bamboo, Pelagic fish
- 157.** The Cuckoo (Koel) lays its eggs in Crow's nest known as:
- (1) Hyper parasitism
- (2) Commensalism
- (3) Mutualism
- (4) Brood parasitism
- 158.** Which one of the following is an example of total stem parasite?
- (1) Santalum
- (2) Rafflesia
- (3) Viscum
- (4) Cuscuta
- 159. Statement-I :** Amazon rain forest is estimated to produce 20% of the total oxygen in the earth's atmosphere.
- Statement-II :** Wild life safari parks are an example of In-situ conservation.
- (1) Both Statement I and Statement II are incorrect.
- (2) Statement I is correct but Statement II is incorrect.
- (3) Statement I is incorrect but Statement II is correct
- (4) Both Statement I and Statement II are correct.

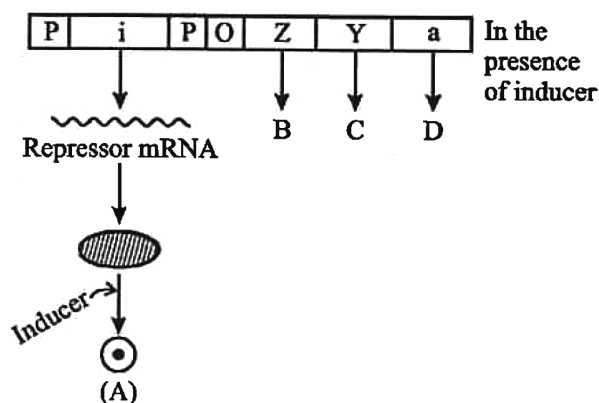
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160. In the following diagram A, B, C and D is respectively.



- (1) A - Inactive repressor, B - β galactosidase, C - Permease, D - Transacetylase
 (2) A - Repressor, B - β galactosidase, C - Permease, D - β galactoside
 (3) A - Repressor, B - β galactosidase, C - Transacetylase, D - Permease
 (4) All of these

161. Point mutation arises due to change in

- (1) single strand of DNA
 (2) single base pair of DNA
 (3) segment of DNA
 (4) double base pair of DNA

162. DNA fingerprinting involves identifying differences in some specific regions in DNA sequence, called as :

- (1) Satellite DNA
 (2) Repetitive DNA
 (3) Single nucleotides
 (4) Polymorphic DNA

163. Sickle cell anaemia is example of all EXCEPT :-

- (1) Gene mutation
 (2) Point mutation
 (3) Base substitution
 (4) Frame-shift mutation

164. Match the following columns :

	Column-A		Column-B
(a)	Expressed sequence tags	(p)	1.4 million locations
(b)	Single nucleotide polymorphisms	(q)	3 billion base pairs
(c)	Less than 2% genome	(r)	Genes Expressed as RNA
(d)	Human genome	(s)	Codes for protein

- (1) (a)-(r), (b)-(p), (c)-(s), (d)-(q)
 (2) (a)-(q), (b)-(r), (c)-(s), (d)-(p)
 (3) (a)-(r), (b)-(s), (c)-(q), (d)-(p)
 (4) (a)-(q), (b)-(p), (c)-(s), (d)-(r)

165. Automated DNA sequencers worked on the principle of a method developed by

- (1) Watson
 (2) Chargaff
 (3) Frederick sanger
 (4) Singer and Nicolson

166. Select the correct sequences of steps in DNA fingerprinting :

- (I) Hybridisation using labelled VNTR probe.
 (II) Isolation of DNA
 (III) Transferring (blotting) of separated DNA fragments to synthetic membranes, such as nitrocellulose or nylon
 (IV) Detection of hybridisation DNA fragments by autoradiography
 (V) Separation of DNA fragments by electrophoresis
 (VI) Digestion of DNA by restriction endonucleases.

- (1) I, V, VI, II, III and IV
 (2) II, VI, V, III, I and IV
 (3) V, I, VI, III, IV and II
 (4) II, I, V, VI, IV, III

167. Mendel proposed which of the following term for hereditary units :-

- (1) Factor (2) Genome
(3) Genetic particle (4) Gene

168. The character which can express only in homozygous condition in case of garden pea is :-

- (a) Constricted pod
(b) Green seed colour
(c) Axial flower
(d) Yellow pod

- (1) A, B & D (2) A, C & D
(3) B, C & D (4) A, B & C

169. Crossing AABB & aabb, the probability of AaBb would be in F_2 generation :-

- (1) 1/16 (2) 2/16
(3) 8/16 (4) 4/16

170. Find out the correct match from column I, II and III of following table :-

I Cross type		II No. of genotypes		III No. of phenotypes	
(A)	Monohybrid	I	9	1	8
(B)	Dihybrid	II	27	2	4
(C)	Trihybrid	III	3	3	2

Choose the correct option :-

- (1) A-III-3, B-I-2, C-II-1
(2) A-II-1, B-III-2, C-I-3
(3) A-I-2, B-III-1, C-II-3
(4) A-III-3, B-II-1, C-I-2

171. Mendel conducted hybridisation experiments for :-

- (1) Seven years (1865-1872)
(2) Seven years (1856-1863)
(3) Seven years (1853-1860)
(4) Fourteen years (1853-1867)

172. How many true breeding lines of garden pea were selected by Mendel ?

- (1) 7
(2) 14
(3) 21
(4) 28

173. Two heterozygous tall plants are crossed then find out percentage of homozygous offspring-

- (1) 25% (2) 50%
(3) 75% (4) 100%

174. How many types of gametes are produced by dihybrid individual?

- (1) Two (2) Three
(3) Eight (4) Four

175. Plant with genotype aaBbccDdEe produces how many types of gametes ?

- (1) 16 (2) 8
(3) 4 (4) 32

176. Which of the following crosses will give tall and dwarf pea plants in same proportions ?

- (1) $TT \times tt$ (2) $Tt \times tt$
(3) $TT \times Tt$ (4) $tt \times tt$

177. When a cross is made between a tall plant with yellow seeds ($TtYy$) and tall plant with green seeds ($Tt yy$), find out the proportions of phenotypes of the offsprings :-

	Proportion of tall and green	Proportion of dwarf and green
(1)	3/8	1/8
(2)	2/8	1/8
(3)	1/8	3/8
(4)	2/8	2/8

178. F_1 generation when resembles :-
 A - either of the two parents
 B - in between of the two parents
 C - with both of the parents
 Choose correct match :-

	A	B	C
(1)	Dominance	Incomplete dominance	Co-dominance
(2)	Dominance	Co-dominance	Incomplete dominance
(3)	Incomplete dominance	Co-dominance	Dominance
(4)	Co-dominance	Dominance	Incomplete dominance

179. Which of the following gamete not obey the law of purity of gametes ?
 (1) RED (2) PINK
 (3) BLACK (4) GREEN
180. A plant shows dominant phenotype and its genotype is to be determined by :-
 (1) Out cross (2) Test cross
 (3) Reciprocal cross (4) Selfing
181. Match and choose correct combination :-

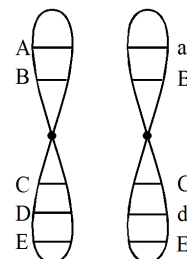
	Column-I		Column-II
A	$Rr \times Rr$	(i)	1 : 1
B	$Rr \times rr$	(ii)	3 : 1
C	$RrYy \times RrYy$	(iii)	1 : 1 : 1 : 1
D	$RrYy \times rryy$	(iv)	9 : 3 : 3 : 1

- (1) A-ii,B-iv,C-iii,D-i
 (2) A-i,B-iii,C-ii,D-iv
 (3) A-iv,B-iii,C-ii,D-i
 (4) A-ii,B-i,C-iv,D-iii

182. Which of the following character was not chosen by Mendel ?

- (1) Pod shape (2) Pod Colour
 (3) Seed Coat Colour (4) Position of flower

183. Given below is a pair of homologous chromosomes :-



How many loci are in heterozygous condition ?

- (1) 2 (2) 3 (3) 5 (4) 10

184. How many total zygotic combinations are possible between a cross $AaBBCc \times AAbbCc$?

- (1) 16 (2) 8 (3) 64 (4) 32

185. Alleles are :

- (1) true breeding homozygotes
 (2) different molecular forms of a gene
 (3) heterozygotes
 (4) different phenotype

SECTION-B (BIOLOGY-II)

186. **Statement-I** : National park, wild life Sanctuaries and biosphere reserves are the example of Ex-situ conservation.

Statement-II : Exploring molecular, genetic and species level diversity for the product of economic importance known as bioprospecting.

- (1) Both Statement I and Statement II are incorrect.
 (2) Statement I is correct but Statement II is incorrect.
 (3) Statement I is incorrect but Statement II is correct
 (4) Both Statement I and Statement II are correct.

187. Statement-I : Government of India in 1980s has introduced the concept of joint forest management (JFM).

Statement-II : Cichlid fish introduced into Lake victoria in east Africa led eventually to the extinction of more than 200 species of Nile perch in the lake.

- (1) Both Statement I and Statement II are incorrect.
- (2) Statement I is correct but Statement II is incorrect.
- (3) Statement I is incorrect but Statement II is correct.
- (4) Both Statement I and Statement II are correct.

188. The correct sequence of plants in a hydrosere is-

- (1) Volvox → Pistia → Scirpus → Hydrilla → Oak
- (2) Volvox → Vallisneria → Pistia → Scirpus → Oak
- (3) Oak → Scirpus → Pistia → Vallisneria → BGA
- (4) BGA → Scirpus → Pistia → Hydrilla → Oak

189. Which one of the following has highest number of producers per unit area ?

- (1) Forest (2) Grassland
- (3) Desert (4) Pond

190. Which one of the following is an example of commensalism.

- A. Orchid - Bee
- B. Orchid - Mango
- C. Clown fish - Sea Anemone
- D. Barnacles - Whale
- E. Sparrow - Seed

- (1) (A), (B), (C) (2) (B), (C), (D)
- (3) (A), (B), (E) (4) (C), (D), (E)

191. Statement-I : Alien species invasion is the most important cause driving animals and plants to extinction.

Statement-II : Many species extinctions in the last 500 years (Steller's Sea cow, Passenger pigeon) where due to Habitat loss and fragmentation.

- (1) Both Statement I and Statement II are incorrect.
- (2) Statement I is correct but Statement II is incorrect.
- (3) Statement I is incorrect but Statement II is correct
- (4) Both Statement I and Statement II are correct.

192. Assertion : The repressor of the operon is synthesized all the time.

Reason : Regulator gene (i) is a constitutive gene.

- (1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
- (2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
- (3) Assertion is True but the Reason is False.
- (4) Both Assertion & Reason are False.

193. The regulation of gene expression in eukaryotes can be exerted at

- I. transcriptional level.
- II. processing level.
- III. transport of mRNA from nucleus of cytoplasm.
- IV. translational level.

Choose the correct combination for the given statements.

- (1) I and II
- (2) II and III
- (3) III and IV
- (4) I, II, III and IV

194. Any Allelic sequence variation has traditionally been described as DNA polymorphism if more than one allele at a locus show :-

- (1) Frequency greater than 0.01
- (2) Frequency less than 0.01
- (3) Frequency greater than 1.0
- (4) Frequency less than 0.01%

195. Which statement is incorrect for VNTR?

- (1) VNTR belongs to a class of satellite DNA referred to as microsatellite
- (2) Copy number of VNTR varies from chromosome to chromosome in an individual
- (3) The size of VNTR varies in size from 0.1 to 20 kb.
- (4) The number of repeat of VNTR show very high degree of polymorphism.

196. Modified allele can produce :-

- (1) Functional enzyme
- (2) Non-functional enzyme
- (3) No enzyme
- (4) All of these

197. Albinism is determined by a recessive gene in humans. The presence of albinism in 50% children born to a couple proves that –

- (1) Both parents are heterozygous
- (2) Father is homozygous normal and mother is albino.
- (3) Both parents are homozygous.
- (4) Father is carrier & mother is albino.

198. In mendelian dihybrid cross, Round, yellow is crossed wrinkled, green seeded pea plant. In F_2 generation, percentage of wrinkled and yellow seed is :-

- (1) 100%
- (2) 56.25%
- (3) 18.75%
- (4) 25%

199. Punnett square represents a dihybrid cross, where round (R) is dominant over wrinkled (r) seeds and yellow (Y) is dominant over green (y) seeds :-

	RY	Ry	rY	ry
RY	F	J	N	R
Ry	G	K	O	S
rY	H	L	P	T
ry	I	M	Q	U

Find out the odd phenotype in following options :-

- (1) G
- (2) H
- (3) L
- (4) K

200. Source of non-parental combinations in mendelian crosses is :-

- (1) Crossing - over
- (2) Dominance
- (3) Independent assortment
- (4) Mutation



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QUESTIONS BREAK-UP CHART FOR MINOR TEST
 FROM CURRENT & PREVIOUS UNITS
 (Session : 2023 - 2024)

ENTHUSE ADV. COURSE : MED

MINOR TEST NO. & DATE	CURRENT UNIT	PREVIOUS UNIT (COMPLETE TOPIC)	
07 10-09-2023	SYLLABUS COVERED FROM 16-08-2023 TO 09-09-2023	P	WAVE OPTICS AND U & D, ERROR
		C	GOC - I
		B	ORIGIN EVOLUTION & DOA

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PHASE - MED

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14-08-2023

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